

ITP- Quiz(A)

Name:

Time: 30 mins

ERP:_____

Total: 20 Marks

Date: 12/3/2024

Q1. You are tasked with designing a C++ program that manages student information. Use a `struct` to represent a student and perform operations on a collection of students. **[10 Marks]**

Each student has the following attributes:

1. **Name** (string): The full name of the student.
2. **Roll Number** (int): The unique roll number of the student.
3. **Marks** (float): The marks obtained by the student in the last exam.

Your program should:

1. Create a dynamically allocated array of structs to store information for up to 50 students.
2. Implement the following functionalities:
 - Add a new student to the system.
 - Search for a student by their roll number and display their details.
 - Update the marks of a student by their roll number.
 - Display all students who scored above a given threshold of marks.

Answer:

Q2. Write a recursive function `reverseString()` that takes a string as input and returns the string reversed. **[5 Marks]**

Examples:

- `reverseString("abc")` returns `"cba"`
- `reverseString("hello")` returns `"olleh"`
- `reverseString("")` returns `" "`

Hint: The reverse of a string can be constructed by appending the last character to the reversed version of the rest of the string.

Answer:

Q3. Short Questions

a. Explain how memory is deallocated for a dynamically created 2D array in C++ using `delete`. [1 Marks]

Answer:

b. Imagine you have a struct `Course` with attributes `courseName (string)` and `credits (int)`, and a struct `Student` with attributes `name (string)`, `rollNumber (int)`, and `enrolledCourses (an array of Course structs)`. Explain how you would store and access the details of all courses a student is enrolled in. [2 Marks]

Answer:

c. Consider the following code. What will happen if you attempt to access an element of the array after deallocating it, and what should we do to avoid that situation? [1 Mark]

```
int** arr = new int*[5];
for (int i = 0; i < 5; i++) {
    arr[i] = new int[10];
}
// Deallocate the array
for (int i = 0; i < 5; i++) {
    delete[] arr[i];
}
delete[] arr;
```

Answer:

d. Consider a `Complex` struct with attributes `real (float)` and `imaginary (float)`. Why can't you directly use the `+` operator to add two `Complex` objects? Write the prototype of the function to overload the `+` operator to add two `Complex` objects. [1 Marks]

Answer: